

Hollandite-Type $\text{VO}_{1.75}(\text{OH})_{0.5}$: Effective Sodium Storage for High-Performance Sodium-Ion Batteries

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Nanosized hollandite-type $\text{VO}_{1.75}(\text{OH})_{0.5}$ is introduced as a novel cathode material for Na-ion batteries. Structural investigation based on X-ray diffraction and Rietveld refinement suggests the presence of numerous vacant sites for Na^+ intercalation in the $\text{VO}_{1.75}(\text{OH})_{0.5}$ structure. All of the possible Na^+ sites and tunnel-type Na^+ diffusion pathways along the c -axis are confirmed by bond-valence-sum analyses. The nanosized hollandite-type $\text{VO}_{1.75}(\text{OH})_{0.5}$ delivers an unexpectedly high specific capacity of $\approx 351 \text{ mAh g}^{-1}$ at 15.5 mA g^{-1} in the voltage range of 1.0–3.7 V (vs Na^+/Na), which agrees well with the results predicted by first-principles calculations. In addition, combined studies using first-principles calculations and several experimental techniques including in situ operando X-ray diffraction and ex situ X-ray absorption spectroscopy confirm that the nanosized hollandite-type $\text{VO}_{1.75}(\text{OH})_{0.5}$ undergoes a single-phase reaction with a capacity retention of 71% over 200 cycles. Furthermore, the open structure and nanosized particles of hollandite-type $\text{VO}_{1.75}(\text{OH})_{0.5}$ contribute to its excellent power capability with 56% of the capacity measured at 0.05 C being delivered at 7 C.

1. Introduction

Lithium-ion batteries (LIBs) have attracted significant attention as high-energy-density rechargeable power sources for electric vehicles and large-scale energy storage applications. Although LIBs have been successfully commercialized in the world

battery market, the expected shortage of lithium resources resulting from the ever-increasing demand for LIBs needs to be addressed. This issue has accelerated the search for alternative power sources such as sodium-ion batteries (SIBs), which are emerging as promising candidates to replace LIBs. Sodium is one of the most plentiful resources on earth; thus, SIBs have been extensively studied because of their low cost as well as their competitiveness in terms of energy density versus LIBs.^[1–5] However, the ionic size of Na^+ (1.02 Å) is larger than that of Li^+ (0.76 Å), which affects the phase stability, transport properties, and inter-phase formation.^[6,7] The low standard redox potential of sodium (−2.71 V vs standard hydrogen electrode (SHE)) compared with that of lithium (−3.04 V vs SHE) is another demerit of SIBs; hence, a high capacity is necessary to compensate for the low operation voltage to approximate the energy density levels of LIBs.^[8]

Thus far, various types of electrode materials have been suggested as cathode materials for SIBs, including $\text{Na}_{2/3}\text{MnO}_2$,^[9] $\text{Na}_{2/3}[\text{M}_x\text{Mn}_{1-x}]\text{O}_2$ (M: transition metals),^[10–13] NaCrO_2 ,^[14] $\alpha\text{-MnO}_2$,^[15] and $\text{Na}_3\text{V}_2\text{PO}_3$.^[16] Vanadium oxides (V_xO_y) is of interest because of the wide range of oxidation states available from +2 (as in VO) to +5 (as in V_2O_5), which affects the structural, electrochemical, and magnetic properties. This property has led to the application of vanadium oxides in catalysts, chemical sensors, and energy storage devices such as rechargeable lithium batteries. The facile insertion of sodium ions into hollandite-type $[2 \times 2]$ tunnel structures in $\alpha\text{-MnO}_2$,^[15] K_xTiO_2 ,^[17] and FeOOH ^[18] has recently been reported. Su et al.^[15] reported that $\alpha\text{-MnO}_2$ delivered a high discharge capacity of 278 mAh g^{-1} in the voltage range of 1–4.3 V for SIBs. However, the capacity retention of $\alpha\text{-MnO}_2$ was unsatisfactory because of the structural instability resulting from repetitive expansion and contraction of the tunnel structures. According to report by Tompsett group,^[19] the $\alpha\text{-MnO}_2$ underwent volume change from 287.3 to 421.5 Å^3 ($\Delta V = 47.7\%$) after 1 mol of Na^+ ion intercalation into the tunnel structure. K_xTiO_2 was shown to be active after carbon coating, enabling successive Na^+ insertion into the tunnel-type pathways with low redox potential. Our prior work utilized hollandite-type FeOOH ,^[7] however, a conversion reaction rather than an intercalation reaction of sodium ions preferentially occurred.

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The chemical formula of hollandite-type vanadium oxides is known to be $A_xV_8O_{16}$ (A = alkaline, alkaline-earth such as $K_2V_8O_{16}$, $K_{1.8}V_8O_{16}$, $Ti_{1.74}V_8O_{16}$, and $Bi_xV_8O_{16}$).^[20–22] Herein, we explore a novel nanosized hollandite-type $VO_{1.75}(OH)_{0.5}$ compound that is similar to the prototype structure as a promising cathode material for SIBs. Using X-ray diffraction (XRD), Rietveld refinement, and bond-valence-sum (BVS) energy maps, we obtained detailed structural information on hollandite-type $VO_{1.75}(OH)_{0.5}$ with large Na^+ diffusion pathways. In particular, we found that in the voltage range between 1.0 and 3.7 V (vs Na^+/Na), $VO_{1.75}(OH)_{0.5}$ delivers an unexpectedly high specific capacity of $\approx 351 \text{ mAh g}^{-1}$ at 15.5 mA g^{-1} , which is comparable to the results predicted by first-principles calculations. Combined studies using several experimental techniques including operando XRD, X-ray absorption nearest edge structure (XANES) spectroscopy, and electrochemical impedance spectroscopy (EIS), and first-principles calculations indicate that $VO_{1.75}(OH)_{0.5}$ undergoes a single-phase reaction during the charging/discharging process accompanying the V^{3+}/V^{4+} redox reaction, resulting in outstanding cycle performance with 71% of its initial capacity retained after 200 cycles. Furthermore, at a rate of 7 C, the specific capacity of $VO_{1.75}(OH)_{0.5}$ is maintained up to $\approx 195 \text{ mAh g}^{-1}$, corresponding to 56% of the capacity measured at 0.05 C.

2. Results and Discussion

The solvothermal product was first observed by scanning electron microscope (SEM) with energy-dispersive X-ray spectroscopy (EDS) to induce the elemental ratio between vanadium and oxygen, which shows the V:O elemental ratio of 1:2.25 (Figure S1, Supporting Information). To analyze the crystal structure and chemical composition of the present material, we combined three methods; namely, XRD, neutron diffraction (ND), and thermal gravimetric analysis (TGA). The original structural model of the $VO_{1.75}(OH)_{0.5}$ from the XRD pattern and the elemental ratio was assumed as the hollandite-type vanadium oxyhydroxide.^[23] Based on structural information on vanadium oxyhydroxide such as space group, atomic position, site occupancy and B_{iso} , we first analyzed the assumed $VO_{1.75}(OH)_{0.5}$ crystal structure using XRD (Figure 1a, Table S1, Supporting Information). To verify the accurate structural information on hydrogen in the structure, we performed Rietveld refinement of the ND data (Figure 1b). ND data were collected using the high-resolution powder diffractometer Echidna at the OPAL research reactor (ANSTO, Lucas Heights) using wavelengths of $1.6215 \text{ \AA}$. All ND peaks of $VO_{1.75}(OH)_{0.5}$ were clearly detected at the ND pattern without any impurities. The refined reliable factors (R-factors) confirm the high accuracy

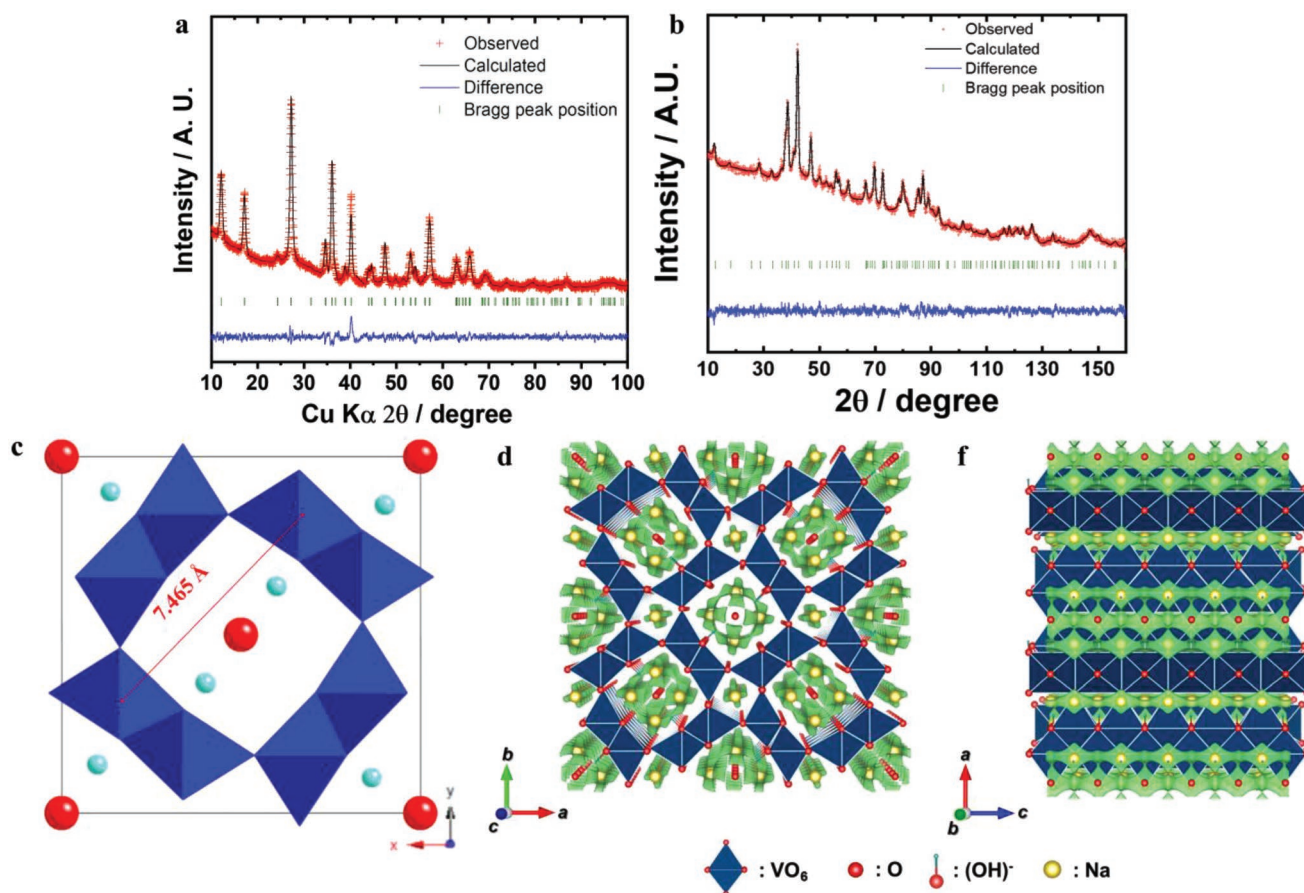
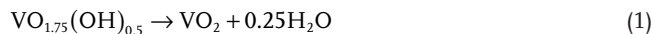


Figure 1. a) Rietveld refinement of XRD data for $VO_{1.75}(OH)_{0.5}$. b) Refined neutron diffraction pattern of $VO_{1.75}(OH)_{0.5}$ ($R_p = 1.18\%$, $R_1 = 4.77\%$, $R_F = 3.92\%$, $\chi^2 = 1.28\%$). c) Tunnel structure of $VO_{1.75}(OH)_{0.5}$. BVS energy maps of $VO_{1.75}(OH)_{0.5}$ in d) bc plane and e) ac plane showing the possible Na ion diffusion paths and positions in the crystal structure.

of our Rietveld refinement results of ND data with the appropriate structural model ($R_p = 1.18\%$, $R_1 = 4.77\%$, $R_F = 3.92\%$, $\chi^2 = 1.28\%$), as described in Table S2 (Supporting Information). Finally, TGA measurement was carried to confirm the results of the above XRD and ND results. $\text{VO}_{1.75}(\text{OH})_{0.5}$ is thermally decomposed, as follow:



From the above reaction, removal of the crystal water corresponds to 5.15 wt% (4.5 g mole^{-1}) weight loss from $\text{VO}_{1.75}(\text{OH})_{0.5}$ ($87.44 \text{ g mole}^{-1}$). According to the TGA data (Figure S2a, Supporting Information), the observed weight loss was about 5.1 wt% to 320°C in air, which agrees with the removal of $0.25\text{H}_2\text{O}$ from $\text{VO}_{1.75}(\text{OH})_{0.5}$. Combination of the above XRD, ND, and TGA data confirms the composition of the present material as $\text{VO}_{1.75}(\text{OH})_{0.5}$. After heating the $\text{VO}_{1.75}(\text{OH})_{0.5}$ at 350°C in air, we checked the resulting XRD pattern and Raman spectroscopy to validate the above reaction. Although the resulting XRD pattern showed low crystalline VO_2 with $C2/m$ space group (PDF# 31-1438) (Figure S2b, Supporting Information), Raman spectrum provides more information on the formation of $\text{VO}_2(\text{B})$ material (Figure S2c, Supporting Information); specifically, 1) $\text{V}_3\text{-O}$ bridging flexural vibration (282 cm^{-1}), 2) the V-O flexural vibration (404 cm^{-1}), 3) $\text{V}_2\text{-O}$ bridging flexural vibration ($400\text{--}600 \text{ cm}^{-1}$), and 4) V-O and V_4O stretching vibration, respectively (691 cm^{-1} and 990 cm^{-1}). Therefore, the structure and chemical composition of $\text{VO}_{1.75}(\text{OH})_{0.5}$ were identified through the combined studies of EDX, XRD, ND, XRD, Raman spectroscopy, and TGA analyses.

Rietveld refinement of the XRD and ND data revealed that $\text{VO}_{1.75}(\text{OH})_{0.5}$ exhibits a typical hollandite-type structure with $I4/m$ space group, with lattice parameters of $a \approx 10.32 \text{ \AA}$ and $c \approx 2.94 \text{ \AA}$ (Figure 1a,b, Tables S1 and S2, Supporting Information). Detailed structural information on $\text{VO}_{1.75}(\text{OH})_{0.5}$, such as the atomic positions, thermal factors, and occupancies, is tabulated in Tables S1 and S2 (Supporting Information). Figure 1c presents the crystal structure of $\text{VO}_{1.75}(\text{OH})_{0.5}$ based on the structural information obtained from the Rietveld refinements of XRD and ND. In the $\text{VO}_{1.75}(\text{OH})_{0.5}$ framework, infinite chains of VO_6 octahedra share their edges with each other, with the chains connected by corner-sharing, resulting in large vacant sites in the structure such as $[2 \times 2]$ -type and $[1 \times 1]$ -type tunnels. The size of the $[2 \times 2]$ -type tunnel for $\text{VO}_{1.75}(\text{OH})_{0.5}$ is $7.465 \text{ \AA}$, which is larger than that of $\alpha\text{-MnO}_2$ ($6.915 \text{ \AA}$).^[24] Thus, Na^+ ions intercalation into the $\text{VO}_{1.75}(\text{OH})_{0.5}$ structure is possible because the large tunnels enable accommodation of the large Na^+ ions. The oxygen ions in the center of the tunnel structure are important during (de)intercalation of sodium ions because the negative O^{2-} ions can reduce the mutual repulsion among Na^+ ions in the hollandite-type structure.^[19]

We constructed bond valence sum (BVS) energy maps to verify the possibility of Na^+ intercalation into the $\text{VO}_{1.75}(\text{OH})_{0.5}$ structure. For the BVS method, the atomic position is assumed to be affected by the difference in the valence of each atom in the structure, allowing prediction of all the possible atomic positions and diffusion paths of Na^+ ions in $\text{VO}_{1.75}(\text{OH})_{0.5}$. As shown in Figure 1d,e, Na^+ ions can occupy not only the

$[2 \times 2]$ -type tunnels but also the $[1 \times 1]$ -type tunnels, with several 1D sodium diffusion pathways along the c -axis in the tunnels. These findings suggest the good power capability of $\text{VO}_{1.75}(\text{OH})_{0.5}$ with fast Na^+ ion diffusion. In addition, two types of sodium sites are present in the $\text{VO}_{1.75}(\text{OH})_{0.5}$ structure with Wyckoff positions and atomic coordinates (x, y, z) of $8h$ ($0.0, 0.125, 0.5$) and $4c$ ($0.5, 0, 0$), which indicates that $\approx 1.5 \text{ mol}$ of Na^+ ions can be theoretically intercalated into $\text{VO}_{1.75}(\text{OH})_{0.5}$ that has high electrical conductivity ($1.2 \times 10^{-4} \text{ S cm}^{-1}$). We also performed the BVS energy map analysis on H^+ ions at $\text{VO}_{1.75}(\text{OH})_{0.5}$ (Figure S3, Supporting Information). The BVS energy map of the diffusion path for H^+ ions is much different from that for the Na^+ ions in $\text{VO}_{1.75}(\text{OH})_{0.5}$. In addition, the possible Na^+ sites at $\text{VO}_{1.75}(\text{OH})_{0.5}$ revealed by the BVS energy map are octahedral sites composed of O^{2-} ions with the ionic size of $\approx 1.4 \text{ \AA}$, which are not suitable for H^+ ions with small ionic size ($\approx 0.2 \text{ \AA}$), because of too small limiting radius ratio between H^+ and O^{2-} ($r^{\text{H}^+}/r^{\text{O}^{2-}} \approx 0.143$).

Transmission electron microscopy (TEM) observation revealed growth of the grain-like particles of $\text{VO}_{1.75}(\text{OH})_{0.5}$ from 50 to 100 nm in size (Figure S4a, Supporting Information). High-resolution transmission electron microscopy (HR-TEM) images in Figure S4b,c (Supporting Information) show the well-formed lattice fringes and growth direction of (002) plane for the grain-like particles. Figure S4d (Supporting Information) presents the particle size distributions of the synthesized materials. Consistent with the TEM data, $\text{VO}_{1.75}(\text{OH})_{0.5}$ was observed to have a narrow distribution in the range of 50 to 150 nm with a high surface area of $83 \text{ m}^2 \text{ g}^{-1}$ (Figure S4d, Supporting Information). These results indicate that $\text{VO}_{1.75}(\text{OH})_{0.5}$ with shortened diffusion pathways is beneficial for enlarging the area participating in the electrochemical reaction.

Using the structural information obtained from Rietveld refinement and the BVS energy maps, we predicted the theoretical properties of $\text{VO}_{1.75}(\text{OH})_{0.5}$ using first-principles calculations. Various Na^+ /vacancy configurations for the $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ compositions ($0 \leq x \leq 1.25$) were generated using the CASM software.^[25] From the formation energy of each configuration, we determined the theoretical redox potential of the $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ electrode using the following equation

$$V = -\frac{E[\text{Na}_{x2}\text{VO}_{1.75}(\text{OH})_{0.5}] - E[\text{Na}_{x1}\text{VO}_{1.75}(\text{OH})_{0.5}] - (x_2 - x_1)E(\text{Na})}{(x_2 - x_1)F} \quad (2)$$

where V is the average redox potential of $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ in the compositional range $x_1 \leq x \leq x_2$, $E[\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}]$ is the formation energy of the most stable configuration of $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ determined by first-principles calculation, $E(\text{Na})$ is the energy of the Na metal, and F is the Faraday constant. As shown in Figure 2a, it was predicted that $\approx 1 \text{ mol}$ of Na^+ ions are reversibly de/intercalated in the $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ structure in the voltage range of 1.0–3.0 V (vs Na^+/Na), and the existence of a stable intermediate phase for the $\text{Na}_{0.5}\text{VO}_{1.75}(\text{OH})_{0.5}$ configuration implies that during the Na^+ ion de/intercalation process, the $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ electrode may undergo a single-phase reaction with a sloppy charge/discharge curve. We first checked the electrochemical activity of $\text{VO}_{1.75}(\text{OH})_{0.5}$ using cyclic voltammogram (CV) of with a sweep rate of 0.5 mV s^{-1}

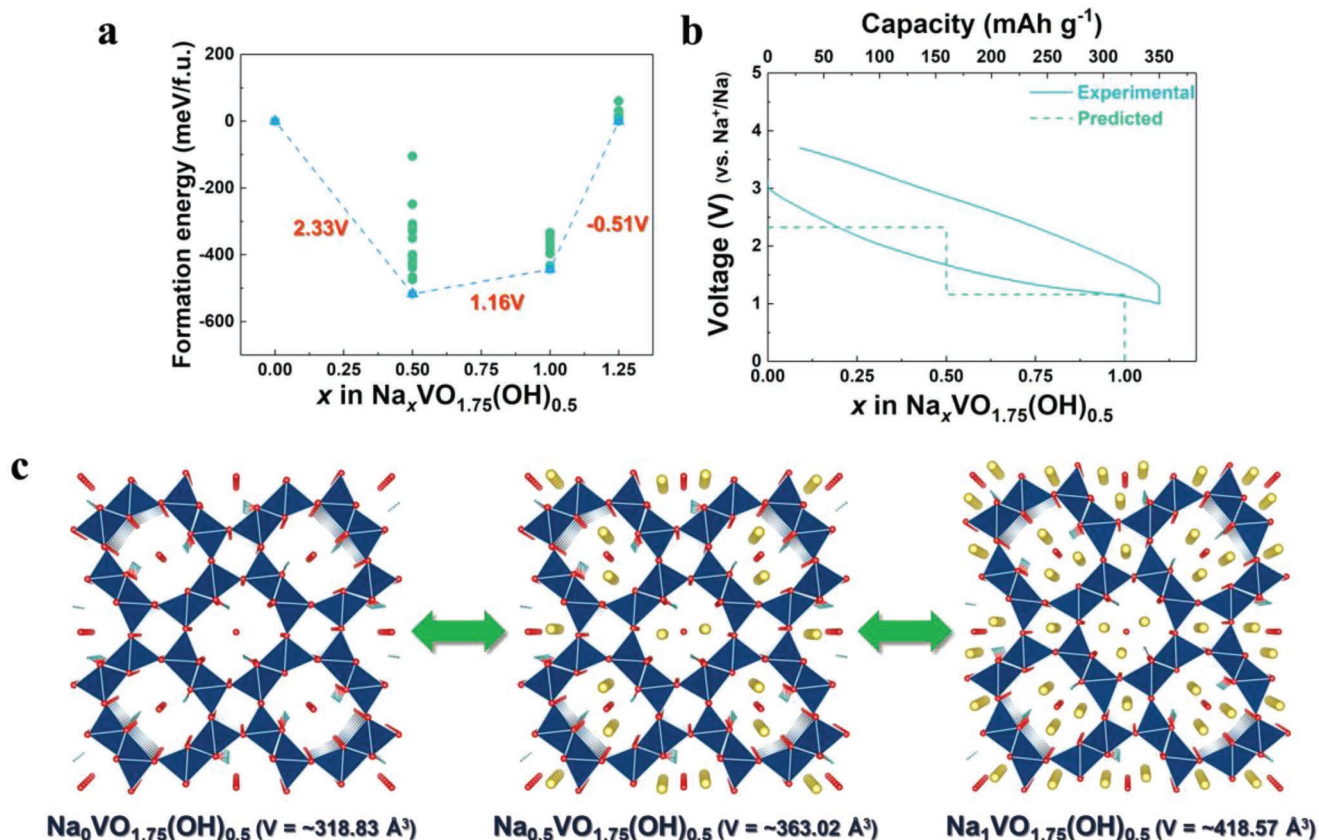


Figure 2. a) Formation energy of $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ ($0 \leq x \leq 1.25$) and b) comparison of experimentally measured charge/discharge curves and predicted voltage curves obtained using first-principles calculations. c) Structural change of $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ ($0 \leq x \leq 1$) predicted using first-principles calculations.

in the voltage range of 0.1–3.7 V (Figure S5, Supporting Information). The appearance of high currents seen in the entire range indicates good electrochemical response of the $\text{VO}_{1.75}(\text{OH})_{0.5}$ electrode in Na cells. A sharp cathodic peak is related to Na^+ ion intercalation into the tunnel structure in the range of 1–3.0 V according to our first-principles calculation (Figure 2a). It seems that the $\text{VO}_{1.75}(\text{OH})_{0.5}$ is active even at low voltage range of 0.1–1 V, of which the reaction may be related to conversion reaction that can induce breakdown the tunnel framework. Since our purpose is to apply this tunnel-type hollandite structure for cathodes, the results mentioned hereafter cover its application to cathode in the range of 1–3.7 V. Based on the CV result, we performed an electrochemical test of the $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ electrode by applying a current rate of 0.05 C ($1 \text{ C} = 310 \text{ mA g}^{-1}$) in the voltage range of 1.0–3.7 V (vs Na^+/Na) and compared the results with the predicted electrochemical properties of $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ (Figure 2b). Our results indicate that $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ can store ≈ 1 mol of Na^+ ions in the structure through the intercalation process. As predicted by the first-principles calculations, $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ exhibited a sloppy charge/discharge curve, which indicates the occurrence of a single-phase reaction accompanying the continuous change in lattice parameters during the Na^+ ion de/intercalation process. Figure 2c shows the predicted crystal structure and volume of $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ as a function of Na^+ ion content.

To confirm the predicted Na^+ ion storage mechanism for $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$, operando XRD (O-XRD) measurements were performed during discharge (reduction) and charge (oxidation) for $\text{VO}_{1.75}(\text{OH})_{0.5}$ (Figure 3a). The main reflections at 27.0° , 34.4° , and 35.7° (2θ), corresponding to (130), (040), and (121) planes, respectively, gradually shifted to lower angles during discharge. To support the XRD data for discharging to 1 V, the resulting XRD pattern was simulated using the FullProf program (Figure 3b). The simulated XRD pattern was consistent with the experimentally determined XRD pattern of $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ discharged to 1 V, with the main (130), (040), and (121) planes shifting to lower 2θ angle. In contrast, the peaks returned to their original Bragg positions on charge, indicating the reversible movement of the major peaks in the XRD pattern. The resulting lattice parameters, which were calculated using the least squares method, also varied reversibly (Figure 3c), coinciding with the movement of the main peaks in the O-XRD data (Figure 3a). In addition, the V K-edge spectra shifted toward V^{3+} from V^{4+} for the fresh state on discharge (1.0 V), and the reduced V^{3+} returned to the original V^{4+} after charging to 3.7 V (Figure 3d). From the above XRD, XANES, and theoretical studies, it is clear that insertion of Na^+ ions into the tunnels induces expansion of the structure, whereas extraction of the Na^+ ions leads to contraction of the tunnels. This work demonstrates the potential of tunnel-structured electrode materials for use as active materials in rechargeable

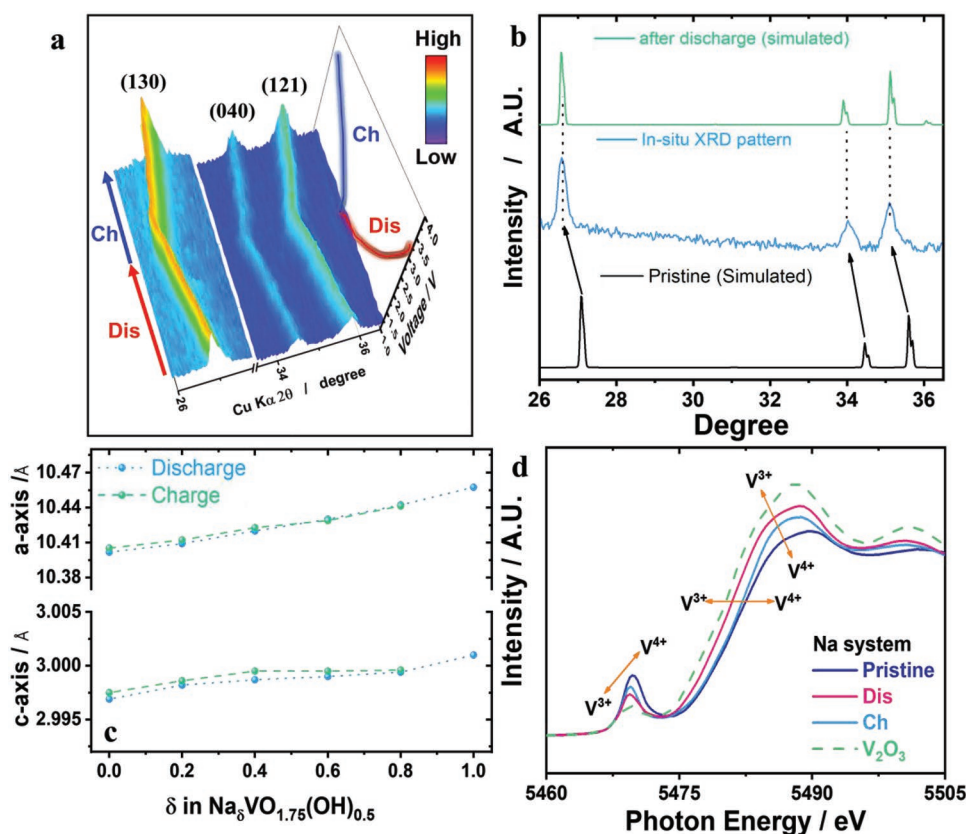


Figure 3. a) Operando XRD pattern of $\text{VO}_{1.75}(\text{OH})_{0.5}$ electrode in a Na cell. b) Magnified XRD pattern of simulated and O-XRD pattern. c) Variation of lattice parameter for $\text{VO}_{1.75}(\text{OH})_{0.5}$. d) V K-edge XANES spectra obtained during discharge and charge of $\text{VO}_{1.75}(\text{OH})_{0.5}$.

sodium batteries with the advantage of high capacity supported by a single-phase reaction and acceptable long-term capacity retention.

To further investigate the electrochemical kinetics of $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$, the contribution of the capacitive behavior to the charging process during Na^+ ion (de)intercalation was evaluated using EIS measurements. EIS spectra were recorded every 0.2 V in a bidirectional discrete voltage scan of $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ (Figure 4a). Quantitative analysis was performed using equivalent electric circuits (Figure 4b); examples of spectra fitting to the equivalent circuit are given in Figure S6 (Supporting Information). The branch of the equivalent circuit which consists of the charge transfer resistance R_d and impedance of diffusion Z_w represents the Faradaic impedance of the reaction with the rate dependent on diffusion (intercalation reaction). The inverse value of the charge-transfer resistance R_d^{-1} is a characteristic of the kinetics of the intercalation (the charge-transfer resistance is inversely proportional to the reaction rate constant). Figure 4c shows the decrease of this parameter during discharge (reduction) but its increase during charge, with a relatively high value of R_d^{-1} at high voltage. Both the inverse charge transfer resistance and the pseudocapacitance variation with the voltage was somewhat dependent on the direction of the voltage scan, which indicated difference of states via which the system passed at same voltage when it was charged and discharged. The pseudocapacitance C_p , which indicates the contribution of a highly reversible surface-limited reaction,^[26]

is the characteristic of the alternative to intercalation. Figure 4d presents the pseudocapacitance normalized by mass to facilitate comparison of the capacitive charging contribution in these cells. It appears that C_p is small for capacitive energy storage and contributes insignificantly for the $\text{VO}_{1.75}(\text{OH})_{0.5}$ electrode. To estimate the order of the contribution, we multiplied the C_p values by the scan rate (0.5 mV s^{-1}) and compared the resulting values with the current obtained from cyclic voltammetry tests at the same scan rate. For $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$, we obtained a capacitive contribution lower than 1.0%, as shown in Table S3 (Supporting Information). Note that the double-layer capacitance was much smaller than the pseudocapacitance and did not contribute significantly to the calculated values. Hence, it is thought that the slight overcapacity, $\approx 351 \text{ mAh g}^{-1}$, at the first discharge for $\text{Na}_1\text{VO}_{1.75}(\text{OH})_{0.5}$ (with a theoretical capacity of 310 mAh g^{-1}) is not associated with the contribution of the pseudocapacitance but is mainly related to the reductive decomposition of the electrolyte in the cell operation range. Given our earlier work using a PC-based electrolyte,^[27] the reductive decomposition is evident in the voltage range of 1.5–0.6 V. The above EIS data suggest that the related electrochemical reaction for $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ is associated with the intercalation of sodium ions into the tunnel-type pathways in the structure.

The power capability of $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ was electrochemically determined by applying various current rates (0.05 C, 0.5 C, 1 C, 2 C, 3 C, 5 C, 7 C). Figure 5a shows that $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ delivered a high discharge capacity of

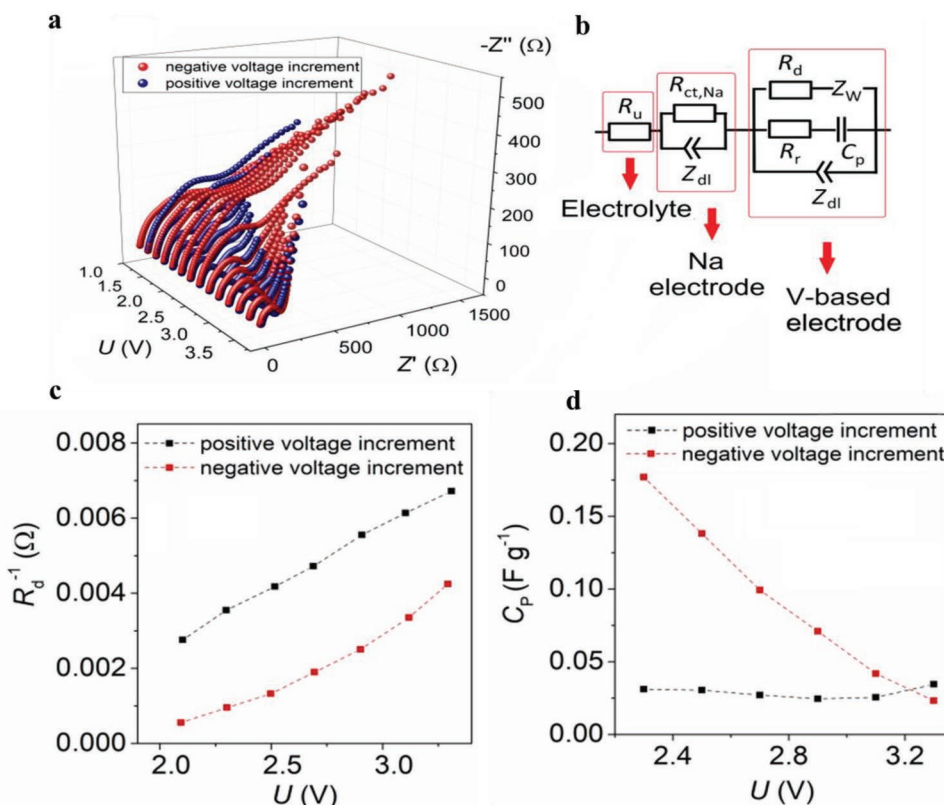


Figure 4. EIS spectra of the coin cells with a) $\text{VO}_{1.75}(\text{OH})_{0.5}$ intercalation host electrodes; b) equivalent electric circuit of appreciably charged battery derived with EISSA software;^[52] c) Dependence of reciprocal charge transfer resistance and d) specific pseudocapacitance on voltage for $\text{VO}_{1.75}(\text{OH})_{0.5}$.

approximately 195 mAh g^{-1} even at a rate of 7 C, corresponding to 56% retention of the capacity measured at 0.05 C. Note that the capacity was recovered when testing was performed at 0.05 C. We supposed that the tunnel-type Na^+ diffusion pathways are repeatedly expanded and contracted by the insertion and extraction of sodium ions over cycles, and the nanosized particles with the shortened diffusion pathway are beneficial for enlarging the area participating in the electrochemical reaction (Figure 5b). Furthermore, it is evident that $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ not only delivered higher discharge/charge capacities but also exhibited outstanding cycle performance as a promising electrode for SIBs. The induction effect on the shortened diffusion path and increased contact area with the electrolyte are likely to facilitate the delivery of the high capacity for $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ and the 71% retention of the capacity ($\approx 250 \text{ mAh g}^{-1}$) over 200 cycles (Figure 5c,d). As shown in Figure 5e,f, the $\text{VO}_{1.75}(\text{OH})_{0.5}$ electrode tested at a current of 1.55 A g^{-1} (5 C) for charge and discharge. The tunnel-structured $\text{VO}_{1.75}(\text{OH})_{0.5}$ electrode presented the first capacities of 219 mAh g^{-1} for discharge and 182 mAh g^{-1} for charge, retaining 84% of the second discharge capacity over cycles. This acceptable high-rate performance is likely ascribed to the stability of the tunnel structure and high electrical conductivity, $1.2 \times 10^{-4} \text{ S cm}^{-1}$, of $\text{VO}_{1.75}(\text{OH})_{0.5}$. Additionally, the nanosized particles that shorten diffusion pathway are also helpful for enlargement of the active area participating in the electrochemical reaction. The volume change of $\text{VO}_{1.75}(\text{OH})_{0.5}$ was compared with $\alpha\text{-MnO}_2$ reported by Tompsett et al.,^[19] of which the $\alpha\text{-MnO}_2$ underwent a large

volume change from 287.3 to $421.5 \text{ \AA}^3$ ($\Delta V = 47.7\%$) after 1 mol of Na^+ ion intercalation into the tunnel structure. As shown in Figure 2a, the $\text{VO}_{1.75}(\text{OH})_{0.5}$ presented volume variation from 318.83 to $418.57 \text{ \AA}^3$ ($\Delta V = 23.8\%$) through 1 mol of Na^+ ion intercalation. From this point of view, it is thought that the less variation in the unit volume of $\text{VO}_{1.75}(\text{OH})_{0.5}$ enables better capacity retention for prolonged cycles (Figure 5d,e).

We measured differential scanning calorimetry (DSC) for the sodiated $\text{Na}_{1.0}\text{VO}_{1.75}(\text{OH})_{0.5}$ and the desodiated $\text{Na}_0\text{VO}_{1.75}(\text{OH})_{0.5}$ obtained after the first discharge and charge, respectively (Figure S7, Supporting Information). Interesting feature is that there were no violent exothermic reactions in the observed temperature range, implying that the present $\text{Na}_0\text{VO}_{1.75}(\text{OH})_{0.5}$ is thermally stable when desodiated and sodiated. The calculated exothermic heats were 0.135 mW mg^{-1} for the sodiated $\text{Na}_{1.0}\text{VO}_{1.75}(\text{OH})_{0.5}$ and 0.278 mW mg^{-1} for the desodiated $\text{Na}_0\text{VO}_{1.75}(\text{OH})_{0.5}$ electrodes. It is likely that the observed small and broad exothermic hills are associated with the removal of crystal water from the $\text{VO}_{1.75}(\text{OH})_{0.5}$ framework, which is evidenced from the TGA curve of $\text{VO}_{1.75}(\text{OH})_{0.5}$ (Figure S2a, Supporting Information). These low heat generation is worth mentioning compared to those violent exothermic reactions of delithiated $\text{Li}_{1-x}\text{CoO}_2$ ^[28] and delithiated Ni-rich compounds.^[29]

We compared the electrochemical performances of the vanadium-based cathode materials for SIBs reported to date (Figure 6a,b), presenting the outstanding properties of the $\text{VO}_{1.75}(\text{OH})_{0.5}$ material.^[6,16,30–44] In particular, the new tunnel-type $\text{VO}_{1.75}(\text{OH})_{0.5}$ shows high specific capacity of 351 mAh g^{-1}

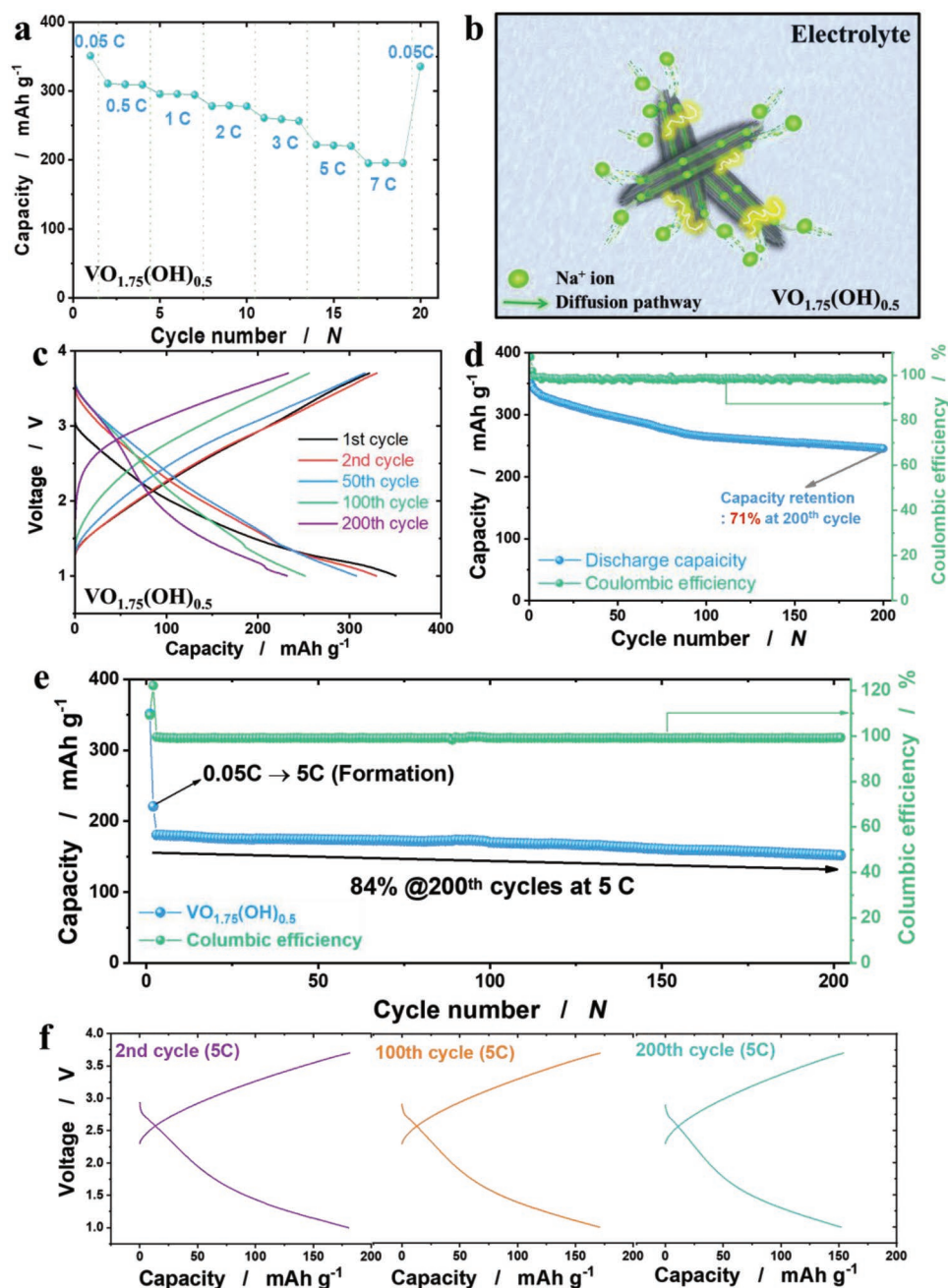


Figure 5. a) Rate capabilities of $\text{VO}_{1.75}(\text{OH})_{0.5}$; b) schematic illustration showing the differences in the diffusion pathway of $\text{VO}_{1.75}(\text{OH})_{0.5}$; c) continuous cycle data of $\text{VO}_{1.75}(\text{OH})_{0.5}$; d) resulting cyclability; e) high-rate cyclability results of $\text{VO}_{1.75}(\text{OH})_{0.5}$ electrode at 5 C rate (1.55 A g^{-1}) with an operational range 1.0–3.7 V; f) selected charge–discharge voltage curves at the 2nd, 100th, and 200th cycle at 5 C rate (1.55 A g^{-1}).

and high energy density of 824 Wh kg^{-1} based on active materials, which highlight outstanding electrochemical performances of the present $\text{VO}_{1.75}(\text{OH})_{0.5}$ as a promising cathode material for SIBs.

3. Conclusion

We report the possibility of using $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ as a new outstanding cathode material for SIBs. Combined studies

using first-principles calculations and various experimental techniques including in situ operando XRD, ex situ XANES, and EIS revealed that large tunnels in the structure enable the insertion of large sodium ions followed by a single-phase reaction with the $\text{V}^{4+}/\text{V}^{3+}$ redox pair. The specific capacity of $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ measured at 0.05 C was $\approx 351 \text{ mAh g}^{-1}$. Furthermore, the nanosized particles of $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ as well as its open structure resulted in excellent power capability with capacity retention of 56% even at 7 C compared with the capacity at 0.05 C. Notably, up to 71% of the initial

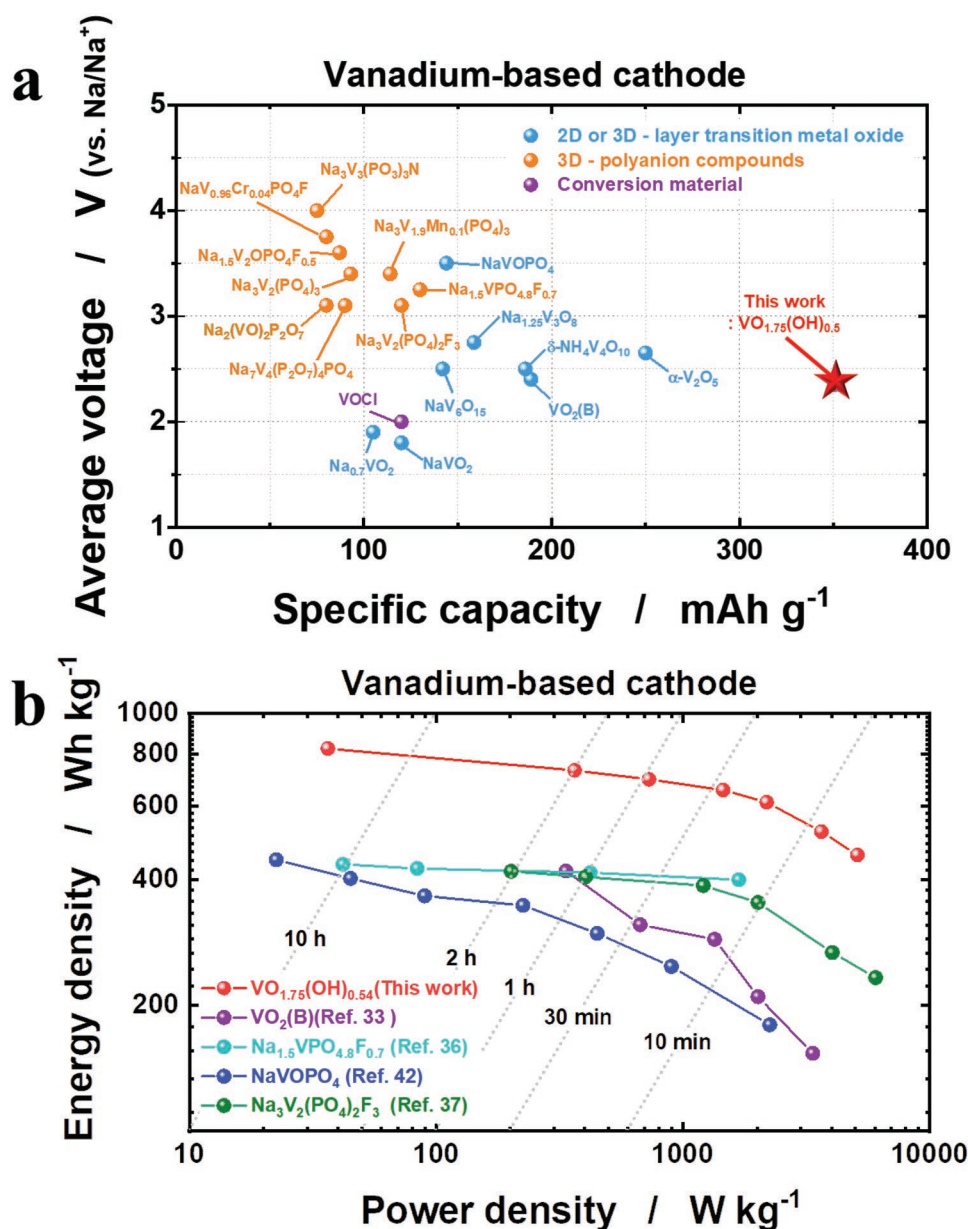


Figure 6. Comparison of a) electrochemical performances and b) Ragone plot among vanadium-based cathode materials for SIBs.

discharge capacity of $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ was maintained for over 200 cycles at 0.05 C, indicating the outstanding cyclability of $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ as a promising cathode material for SIBs. We believe that our novel approach will provide insight for the discovery and understanding of novel electrode materials for not only SIBs but also other rechargeable batteries.

4. Experimental Section

The $\text{VO}_{1.75}(\text{OH})_{0.5}$ in this study was synthesized using a solvothermal method. First, vanadium oxytrichloride (Aldrich) was slowly added to benzyl alcohol solution in a vial in an Ar-filled glove box. The contents of the vial were then stirred at room temperature until the red solution turned green and finally blue. Next, the solution was autoclaved at 150 °C for 1 d under autogenous pressure. The resulting black powders

were filtered and washed with ethanol and then dried at 60 °C overnight in the oven.

XRD (6 kW, X'Pert, PANalytical) using Cu K α radiation was employed to identify the crystalline phases of the products in the 2θ range between 10° and 100° using a step size of 0.03°. The obtained XRD data were refined using the FullProf Rietveld program. ND data were collected using the high-resolution powder diffractometer Echidna at the OPAL research reactor (ANSTO, Lucas Heights) using wavelengths of 1.6215 Å. The as-prepared powder products were also examined using field-emission scanning electron microscopy (FE-SEM; JXA-8100, JEOL) and HR-TEM (JEM-3010, JEOL). For thermogravimetric analysis (TGA, SDT, STA instruments, USA), the loaded pristine samples (≈ 13 mg) were heated from room temperature to 500 °C at a heating and cooling rate of 5 °C min⁻¹. The particle size distribution was determined into the disposable cell using deionized water with dispersion medium by dynamic light scattering (ELS-Z2000, Otsuka Electronics Co., Ltd.). Raman spectroscopy (Renishaw, inVia Raman Microscope) was used

to confirm the presenting bonds between V and O for dehydrated $\text{VO}_{1.75}(\text{OH})_{0.5}$.

The produced materials were blended with electroconducting carbon (Super P and Denka black, 1:1 by weight) and polyvinylidene fluoride with a weight ratio of 8:1:1 in *N*-methyl-2-pyrrolidone (NMP). The obtained slurry was applied onto aluminum foil and then dried at 80 °C overnight under vacuum. The typical electrode weight was $\approx 3.0 \text{ mg cm}^{-2}$. Electrochemical cell tests were conducted after assembling R2032 coin-type cells with Na metal as the negative electrode in 0.5 M NaPF_6 solution in propylene carbonate (PC) and fluoroethylene carbonate (98:2 by volume). All the cells were assembled in an Ar-filled glove box. The cells were cycled between 1.0 and 3.7 V at a current density of 15.5 mA g^{-1} at 25 °C. A Metrohm Autolab PGSTAT204 potentiostat with FRA32M module was used for impedance spectra acquisition. The two-electrode setup was applied for coin cells. The cell voltage was varied between 1.0 and 3.7 V: from high voltage to low voltage and then from low voltage to high voltage. EIS spectra were recorded with 0.2 V step and 30 min conditioning time. Frequency range was 100 kHz–0.01 Hz, and AC amplitude of 10 mV was applied. Differential scanning calorimetry (DSC, Pyris 1 Calorimeter, Perkin-Elmer) was used to understand thermal behavior of electrochemically de-/sodiated electrodes (3–5 mg). Details of sample preparation and measurement conditions are described in our earlier report.^[45]

Operando XRD was used to investigate the structural evolution during (de)sodiation and charging, and *ex situ* XANES spectroscopy at the 8C Nano XAFS beamline at Pohang Accelerator Laboratory (PAL, Pohang, South Korea) was performed to observe the variation in the oxidation state of vanadium.

Density functional theory (DFT) calculations were performed using the Vienna Ab initio Simulation Package (VASP).^[46] The projector-augmented wave (PAW) pseudopotential with a plane-wave basis set was used.^[47] Perdew–Burke–Ernzerhof (PBE) parametrization of the generalized gradient approximation (GGA) was used for the exchange–correlation function.^[48] The GGA+U method was adopted to address the localization of the d-orbital in the V ions,^[49] with a U value of 4.2 eV, which was previously determined for V-based electrode materials for SIBs.^[50,51] All the calculations were performed with an energy cutoff of 500 eV until the remaining force in the system converged to less than 0.05 eV Å^{-1} per unit cell. A $3 \times 3 \times 10$ k-point grid was used to calculate a $1 \times 1 \times 1$ supercell structure of $\text{VO}_{1.75}(\text{OH})_{0.5}$. The phase stability of $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$ was investigated by generating all the Na^+ ion configurations corresponding to specific Na^+ ion contents and calculating the formation energies. Cluster-assisted statistical mechanics (CASM)^[25] software was used to generate all of the Na/vacancy configurations for each composition; this step was followed by full DFT calculations for a maximum of 30 configurations with the lowest electrostatic energy for each composition to obtain a convex hull plot for $\text{Na}_x\text{VO}_{1.75}(\text{OH})_{0.5}$.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

first-principles calculations, Hollandite, intercalation, sodium batteries, vanadium oxyhydroxide

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- [1] Y.-K. Sun, S.-T. Myung, B.-C. Park, J. Prakash, I. Belharouak, K. Amine, *Nat. Mater.* **2009**, *8*, 320.
- [2] M. Kunduraci, J. F. Al-Sharab, G. G. Amatucci, *Chem. Mater.* **2006**, *18*, 3585.
- [3] J. H. Jo, J. U. Choi, A. Konarov, H. Yashiro, S. Yuan, L. Shi, Y.-K. Sun, S.-T. Myung, *Adv. Funct. Mater.* **2018**, *28*, 1705968.
- [4] H. Zhang, X. P. Gao, G. R. Li, T. Y. Yan, H. Y. Zhu, *Electrochim. Acta* **2008**, *53*, 7061.
- [5] Z. Chen, J. R. Dahn, *Electrochim. Acta* **2004**, *49*, 1079.
- [6] N. B. Mahadi, J. S. Park, J. H. Park, K. Y. Chung, Y. K. Sun, S. T. Myung, *J. Power Sources* **2016**, *326*, 522.
- [7] M. K. Cho, J. H. Jo, J. U. Choi, J. Kim, H. Yashiro, S. Yuan, L. Shi, Y.-K. Sun, S.-T. Myung, *Nano Energy* **2017**, *41*, 687.
- [8] J. Paramudita, D. Sehwat, D. Goonetilleke, N. Sharma, *Adv. Energy Mater.* **2017**, *7*, 1602911.
- [9] S. Kumakura, Y. Tahara, K. Kubota, K. Chihara, S. Komaba, *Angew. Chem.* **2016**, *128*, 12952; *Angew. Chem., Int. Ed.* **2016**, *55*, 12760.
- [10] D.-H. Lee, J. Xu, Y.-S. Meng, *Phys. Chem. Chem. Phys.* **2013**, *15*, 3304.
- [11] N. Yabuuchi, M. Kajiyama, J. Iwatate, H. Nishikawa, S. Hitomi, R. Okuyama, R. Usui, Y. Yamada, S. Komaba, *Nat. Mater.* **2012**, *11*, 512.
- [12] F. Tournadre, L. Croguennec, I. Saadoune, D. Carlier, Y. Shao-Horn, P. Willmann, C. Delmas, *J. Solid State Chem.* **2004**, *177*, 2790.
- [13] J. U. Choi, Y. J. Park, J. H. Jo, L.-Y. Kuo, P. Kaghazchi, S.-T. Myung, *ACS Appl. Mater. Interfaces* **2018**, *10*, 40978.
- [14] C. Y. Yu, J. S. Park, H. G. Jung, K. Y. Chung, D. Aurbach, Y.-K. Sun, S.-T. Myung, *Energy Environ. Sci.* **2015**, *8*, 2019.
- [15] D. Su, H.-J. Ahn, G. Wang, *J. Mater. Chem. A* **2013**, *1*, 4845.
- [16] J.-S. Park, J. Kim, J. H. Jo, S.-T. Myung, *J. Mater. Chem. A* **2018**, *6*, 16627.
- [17] Q. Zhang, Y. Wei, H. Yang, D. Su, Y. Ma, H. Li, T. Zhai, *ACS Appl. Mater. Interfaces* **2017**, *9*, 7009.
- [18] E. Zhang, B. Wang, X. Yu, J. Zhu, L. Wang, B. Lu, *Energy Storage Mater.* **2017**, *8*, 147.
- [19] D. A. Tompsett, M. S. Islam, *Chem. Mater.* **2013**, *25*, 2515.
- [20] W. Abriel, F. Rau, K. J. Range, *Mater. Res. Bull.* **1979**, *14*, 1463.
- [21] F. Abraham, O. Mentre, *J. Solid State Chem.* **1994**, *109*, 127.
- [22] J. H. Jo, Y.-K. Sun, S.-T. Myung, *J. Mater. Chem. A* **2017**, *5*, 8367.
- [23] I. Djerdj, D. Sheptyakov, F. Gozzo, D. Arcon, R. Nesper, M. Niederberger, *J. Am. Chem. Soc.* **2008**, *130*, 11364.
- [24] M. H. Alfaruqi, J. Gim, S. Kim, J. Song, J. Jo, S. Kim, V. Mathew, J. Kim, *J. Power Sources* **2015**, *288*, 320.

- [25] A. V. D. Ven, J. C. Thomas, Q. Xu, J. Bhattacharya, *Math. Comput. Simul.* **2010**, *80*, 1393.
- [26] G. A. Ragoisha, *Electroanalysis* **2015**, *27*, 855.
- [27] S. Komaba, T. Ishikawa, N. Yabuuchi, W. Murata, A. Ito, Y. Ohsawa, *ACS Appl. Mater. Interfaces* **2011**, *3*, 4165.
- [28] K.-S. Lee, S.-T. Myung, D.-W. Kim, Y.-K. Sun, *J. Power Sources* **2011**, *196*, 6974.
- [29] H.-H. Sun, W. Choi, J. K. Lee, I.-H. Oh, H.-G. Jung, *J. Power Sources* **2015**, *275*, 877.
- [30] T. McQueen, P. Stephenes, Q. Huang, T. E. Klimczuk, F. Ronning, R. Cava, *Phys. Rev. Lett.* **2008**, *101*, 166402.
- [31] M. Guignard, C. Didier, J. Darriet, P. Bordet, E. Elkaim, C. Delmas, *Nat. Mater.* **2013**, *12*, 74.
- [32] S. Tepavcevic, H. Xiong, V. R. Stamenkovic, X. Zuo, M. Balasubramanian, V. B. Prakapenka, C. S. Johnson, T. Rajh, *ACS Nano* **2012**, *6*, 530.
- [33] H. Zhuo, X. Yang, A. Tang, Z. Liu, S. Gamboa, P. J. Sebastian, *J. Power Sources* **2006**, *160*, 698.
- [34] F. Sauvage, E. Quarez, J.-M. Tarascon, E. Baudrin, *Solid State Sci.* **2006**, *8*, 1215.
- [35] Y.-U. Park, D.-H. Seo, H.-S. Kwon, B. Kim, J. Kim, H. Kim, I. Kim, H.-I. Yoo, K. Kang, *J. Am. Chem. Soc.* **2013**, *135*, 13870.
- [36] Q. Liu, D. Wang, X. Yang, N. Chen, C. Wang, X. Bie, Y. Wei, G. Chen, F. Du, *J. Mater. Chem. A* **2015**, *3*, 21478.
- [37] P. Barpanda, G. Liu, M. Avdeev, A. Yamada, *ChemElectroChem* **2014**, *1*, 1488.
- [38] S. Y. Lim, H. Kim, J. Chung, J. H. Lee, B. G. Kim, J.-J. Choi, K. Y. Chung, W. Cho, S.-J. Kim, W. A. Goddard III, Y. Jung, J.-W. Choi, *Proc. Natl. Acad. Sci. USA* **2014**, *111*, 599.
- [39] M. Reynaud, A. Wizner, N. A. Katcho, L. C. Loaiza, M. Galceran, J. Carrasco, T. Rojo, M. Armand, M. C. Cabanas, *Electrochem. Commun.* **2017**, *84*, 14.
- [40] Y. Fang, Q. Liu, L. Xiao, Y. Rong, Y. Liu, Z. Chen, X. Ai, Y. Cao, H. Yang, J. Xie, C. Sun, X. Zhang, B. Aoun, X. Xing, X. Xiao, Y. Ren, *Chem* **2018**, *4*, 1.
- [41] Y. Dong, S. Li, K. Zhao, C. Han, W. Chen, B. Wang, L. Wang, B. Xu, Q. Wei, L. Zhang, X. Xu, L. Mai, *Energy Environ. Sci.* **2015**, *8*, 1267.
- [42] H. Liu, H. Zhou, L. Chen, Z. Tang, W. Yang, *J. Power Sources* **2011**, *196*, 814.
- [43] H. Fei, X. Liu, Y. Lin, M. Wei, *J. Colloid Interface Sci.* **2014**, *428*, 73.
- [44] P. Gao, C. Wall, L. Zhang, M. A. Redd, M. Fichtner, *Electrochem. Commun.* **2015**, *60*, 180.
- [45] C.-H. Jo, J.-H. Jo, H. Yashiro, S.-J. Kim, Y.-K. Sun, S.-T. Myung, *Adv. Energy Mater.* **2018**, *8*, 1702942.
- [46] G. Kresse, J. Furthmüller, *Comput. Mater. Sci.* **1996**, *6*, 15.
- [47] P. E. Blöchl, *Phys. Rev. B* **1994**, *50*, 17953.
- [48] J. P. Perdew, K. Burke, M. Ernzerhof, *Phys. Rev. Lett.* **1996**, *77*, 3865.
- [49] V. I. Anisimov, F. Aryasetiawan, A. I. Lichtenstein, *J. Phys.: Condens. Matter* **1997**, *9*, 767.
- [50] S. Y. Lim, H. Kim, R. A. Shakoor, Y. Jung, J. W. Choi, *J. Electrochem. Soc.* **2012**, *159*, A1393.
- [51] J. Kim, G. Yoon, H. Kim, Y.-U. Park, K. Kang, *Chem. Mater.* **2018**, *30*, 3683.
- [52] A. S. Bondarenko, G. A. Ragoisha, in *Progress in Chemometrics Research* (Ed: A. L. Pomerantsev), Nova Science Publishers, New York **2005**, pp. 89–102.